

Physics 513 QFT: HW 8

Pavel Okun

Department of Chemistry, University of Michigan, Ann Arbor USA

(Dated: Thursday 25th December, 2025)

CONTENTS

1. 1
 - 1.1. Proving Eq. (1)
 - 1.2. Proving Eq. (2)
 - 1.3. Proving the Gordon Identity
2. 2
 - 2.1. γ^μ
 - 2.2. $S^{\mu\nu}$
 - 2.3. $i\gamma_5$
 - 2.4. $\gamma^\mu\gamma_5$
 - 2.5. $i\gamma_5 S^{\mu\nu}$

Using the identity

$$2AB = AB + AB = AB + BA + AB - BA = \{A, B\} + [A, B] \quad (9)$$

yields

$$2\gamma^\mu \not{p} = \{\gamma^\mu, \not{p}\} + [\gamma^\mu, \not{p}] = 2p^\mu I - 4iS^{\mu\nu}p_\nu. \quad (10)$$

Suppressing the identity matrix I yields Eq. 2

$$\boxed{2\gamma^\mu \not{p} = \{\gamma^\mu, \not{p}\} + [\gamma^\mu, \not{p}] = 2p^\mu - 4iS^{\mu\nu}p_\nu.} \quad (11)$$

QED.

1. 1

1.1. Proving Eq. (1)

As the HW suggests I begin by proving Eq. (1) in the HW

$$\gamma^\mu \not{p} = \frac{\{\gamma^\mu, \not{p}\} + [\gamma^\mu, \not{p}]}{2} = -p^\mu - 2iS^{\mu\nu}p_\nu. \quad (1)$$

This equation has a typo. It is off by a minus sign ($-p^\mu \rightarrow p^\mu$). I fix this typo and reorganize

$$2\gamma^\mu \not{p} = \{\gamma^\mu, \not{p}\} + [\gamma^\mu, \not{p}] = 2p^\mu - 4iS^{\mu\nu}p_\nu. \quad (2)$$

I use the notation and definitions in my QED notes, which are consistent with Schwartz.

$$\not{p} = \gamma^\nu p_\nu. \quad (3)$$

$$\{\gamma^\mu, \not{p}\} + [\gamma^\mu, \not{p}] = \{\gamma^\mu, \gamma^\nu p_\nu\} + [\gamma^\mu, \gamma^\nu p_\nu] \quad (4)$$

$$\{\gamma^\mu, \not{p}\} + [\gamma^\mu, \not{p}] = \{\gamma^\mu, \gamma^\nu\}p_\nu + [\gamma^\mu, \gamma^\nu]p_\nu \quad (5)$$

$$-4iS^{\mu\nu} = [\gamma^\mu, \gamma^\nu], \quad \{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}I \quad (6)$$

I is the 4×4 identity matrix and η is the Minkowski metric: $\eta^{\alpha\beta} = \text{diag}(1, -1, -1, -1)$ in Cartesian coordinates.

$$\{\gamma^\mu, \not{p}\} + [\gamma^\mu, \not{p}] = 2\eta^{\mu\nu}p_\nu I - 4iS^{\mu\nu}p_\nu \quad (7)$$

$$\{\gamma^\mu, \not{p}\} + [\gamma^\mu, \not{p}] = 2p^\mu I - 4iS^{\mu\nu}p_\nu \quad (8)$$

1.2. Proving Eq. (2)

Now I prove Eq. (2) in the HW

$$\not{p}'\gamma^\mu = \frac{\{\gamma^\mu, \not{p}'\} - [\gamma^\mu, \not{p}']}{2} = -p'^\mu + 2iS^{\mu\nu}p'_\nu. \quad (12)$$

I make this look nicer by taking $\vec{p}' \rightarrow \vec{p}$. I also fix the same typo again ($-p^\mu \rightarrow p^\mu$) and rearrange

$$2\not{p}\gamma^\mu = \{\gamma^\mu, \not{p}\} - [\gamma^\mu, \not{p}] = 2p^\mu + 4iS^{\mu\nu}p_\nu. \quad (13)$$

The identity

$$2AB = AB + AB = AB + BA - BA + AB = \{A, B\} - (BA - AB) = \{A, B\} - [B, A] = \{B, A\} - [B, A] \quad (14)$$

implies

$$2\not{p}\gamma^\mu = \{\gamma^\mu, \not{p}\} - [\gamma^\mu, \not{p}]. \quad (15)$$

$$2\not{p}\gamma^\mu = \{\gamma^\mu, \not{p}\} - [\gamma^\mu, \not{p}] = \{\gamma^\mu, \gamma^\nu p_\nu\} - [\gamma^\mu, \gamma^\nu p_\nu] \quad (16)$$

$$2\not{p}\gamma^\mu = \{\gamma^\mu, \not{p}\} - [\gamma^\mu, \not{p}] = \{\gamma^\mu, \gamma^\nu\}p_\nu - [\gamma^\mu, \gamma^\nu]p_\nu \quad (17)$$

$$2\not{p}\gamma^\mu = \{\gamma^\mu, \not{p}\} - [\gamma^\mu, \not{p}] = 2\eta^{\mu\nu}p_\nu I + 4iS^{\mu\nu}p_\nu \quad (18)$$

$$\boxed{2\not{p}\gamma^\mu = \{\gamma^\mu, \not{p}\} - [\gamma^\mu, \not{p}] = 2p^\mu + 4iS^{\mu\nu}p_\nu} \quad (19)$$

Thus I proved Eq. 13, QED.

1.3. Proving the Gordon Identity

The Gordon identity is

$$\bar{u}(\vec{p}')\gamma^\mu u(\vec{p}) = \bar{u}(\vec{p}')\frac{(\vec{p} + \vec{p}')^\mu + iS^{\mu\nu}(\vec{p}' - \vec{p})_\nu}{2m}u(\vec{p}). \quad (20)$$

I relabel $\vec{p}' \rightarrow \vec{k}$. There is another typo in this problem. If you compare with Eq. (11.92) in Schwartz you will see that we need $S^{\mu\nu} \rightarrow \sigma^{\mu\nu}$ where

$$\sigma^{\mu\nu} = \frac{i}{2}[\gamma^\mu, \gamma^\nu] = 2S^{\mu\nu}. \quad (21)$$

Now the Gordon identity is

$$2m \bar{u}(\vec{k})\gamma^\mu u(\vec{p}) = \bar{u}(\vec{k})[(\vec{p} + \vec{k})^\mu + 2iS^{\mu\nu}(\vec{k} - \vec{p})_\nu]u(\vec{p}). \quad (22)$$

I prove this equation.

$$2m \bar{u}(\vec{k})\gamma^\mu u(\vec{p}) = \bar{u}(\vec{k})[p^\mu + k^\mu + 2iS^{\mu\nu}(k_\nu - p_\nu)]u(\vec{p}) \quad (23)$$

$$2m \bar{u}(\vec{k})\gamma^\mu u(\vec{p}) = \bar{u}(\vec{k})(p^\mu + k^\mu + 2iS^{\mu\nu}k_\nu - 2iS^{\mu\nu}p_\nu)u(\vec{p}) \quad (24)$$

$$4m \bar{u}(\vec{k})\gamma^\mu u(\vec{p}) = \bar{u}(\vec{k})(2p^\mu - 4iS^{\mu\nu}p_\nu + 2k^\mu + 4iS^{\mu\nu}k_\nu)u(\vec{p}) \quad (25)$$

Plugging in Eqs. 11 & 19

$$2\gamma^\mu \not{p} = 2p^\mu - 4iS^{\mu\nu}p_\nu, \quad 2\not{k}\gamma^\mu = 2k^\mu + 4iS^{\mu\nu}k_\nu, \quad (26)$$

yields

$$2m \bar{u}(\vec{k})\gamma^\mu u(\vec{p}) = \bar{u}(\vec{k})(\gamma^\mu \not{p} + \not{k}\gamma^\mu)u(\vec{p}). \quad (27)$$

$$2m \bar{u}(\vec{k})\gamma^\mu u(\vec{p}) = \bar{u}(\vec{k})\gamma^\mu \not{p}u(\vec{p}) + \bar{u}(\vec{k})\not{k}\gamma^\mu u(\vec{p}). \quad (28)$$

Everything in this problem is classical. I take the classical single mode free spinor field, Eq. (11.13) in Schwartz,

$$\psi(\vec{x}) = u(\vec{p})e^{-i\vec{p}\cdot\vec{x}}, \quad (29)$$

and Plug it into the Dirac equation.

$$(i\not{\partial} - m)\psi(\vec{x}) = 0 \quad (30)$$

$$i\not{\partial}\psi(\vec{x}) - m\psi(\vec{x}) = 0 \quad (31)$$

$$i\not{\partial}u(\vec{p})e^{-i\vec{p}\cdot\vec{x}} - mu(\vec{p})e^{-i\vec{p}\cdot\vec{x}} = 0 \quad (32)$$

$$(i\not{\partial}e^{-i\vec{p}\cdot\vec{x}} - me^{-i\vec{p}\cdot\vec{x}})u(\vec{p}) = 0 \quad (33)$$

$$(i\gamma^\mu\partial_\mu e^{-i\vec{p}\cdot\vec{x}} - me^{-i\vec{p}\cdot\vec{x}})u(\vec{p}) = 0 \quad (34)$$

$$\begin{aligned} \partial_\mu e^{-i\vec{p}\cdot\vec{x}} &= \partial_\mu \exp(-ip_\beta x^\beta) = -ip_\beta e^{-i\vec{p}\cdot\vec{x}} \partial_\mu x^\beta = \\ &= -ip_\beta e^{-i\vec{p}\cdot\vec{x}} \frac{\partial x^\beta}{\partial x^\mu} = -ip_\beta e^{-i\vec{p}\cdot\vec{x}} \delta_\mu^\beta = -ie^{-i\vec{p}\cdot\vec{x}} p_\mu \end{aligned} \quad (35)$$

$$(\gamma^\mu p_\mu - m)e^{-i\vec{p}\cdot\vec{x}}u(\vec{p}) = 0 \quad (36)$$

$$(\not{p} - m)u(\vec{p}) = 0 \quad (37)$$

$$\boxed{\not{p}u(\vec{p}) = mu(\vec{p})} \quad (38)$$

I take the conjugate of this equation.

$$\overline{\not{p}u(\vec{p})} = m\bar{u}(\vec{p}) \quad (39)$$

$$\overline{\gamma^\mu p_\mu u(\vec{p})} = m\bar{u}(\vec{p}) \quad (40)$$

$$p_\mu \overline{\gamma^\mu u(\vec{p})} = m\bar{u}(\vec{p}) \quad (41)$$

$\gamma^\mu u$ is a Dirac spinor and I know how to take the conjugate of a Dirac spinor

$$\bar{\psi} = \psi^\dagger \gamma^0. \quad (42)$$

$$\overline{\gamma^\mu u} = (\gamma^\mu u)^\dagger \gamma^0 = u^\dagger \gamma^{\mu\dagger} \gamma^0 \quad (43)$$

I use two Dirac matrix properties

$$I = (\gamma^0)^2, \quad \gamma^\mu = \gamma^0 \gamma^{\mu\dagger} \gamma^0. \quad (44)$$

$$\overline{\gamma^\mu u} = u^\dagger \gamma^0 \gamma^0 \gamma^{\mu\dagger} \gamma^0 = \bar{u} \gamma^0 \gamma^{\mu\dagger} \gamma^0 = \bar{u} \gamma^\mu \quad (45)$$

$$p_\mu \bar{u}(\vec{p}) \gamma^\mu = m\bar{u}(\vec{p}) \quad (46)$$

$$\bar{u}(\vec{p}) \gamma^\mu p_\mu = m\bar{u}(\vec{p}) \quad (47)$$

$$\boxed{\bar{u}(\vec{p}) \not{p} = m\bar{u}(\vec{p})} \quad (48)$$

I plug Eqs. 38 & 48 into Eq. 28.

$$2m \bar{u}(\vec{k})\gamma^\mu u(\vec{p}) = \bar{u}(\vec{k})\gamma^\mu \not{p}u(\vec{p}) + \bar{u}(\vec{k})\not{k}\gamma^\mu u(\vec{p}) \quad (49)$$

$$2m \bar{u}(\vec{k})\gamma^\mu u(\vec{p}) = \bar{u}(\vec{k})\gamma^\mu mu(\vec{p}) + m\bar{u}(\vec{k})\not{k}\gamma^\mu u(\vec{p}) \quad (50)$$

$$2m \bar{u}(\vec{k})\gamma^\mu u(\vec{p}) = 2m\bar{u}(\vec{k})\gamma^\mu u(\vec{p}) \quad (51)$$

$$1 = 1 \quad (52)$$

I have proved the Gordon identity. QED.

2. 2

2.1. γ^μ

$$\gamma^\mu = \bar{\gamma}^\mu = \gamma^0 \gamma^{\mu\dagger} \gamma^0 \quad (53)$$

I do the $\mu = 0$ case.

$$\bar{\gamma}^0 = \gamma^0 \gamma^{0\dagger} \gamma^0 \quad (54)$$

$$\gamma^0 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \implies \gamma^0 = \gamma^{0\dagger} \quad (55)$$

The Dirac matrices are 4×4 so each element represents a 2×2 matrix.

$$\bar{\gamma}^0 = \gamma^0 \gamma^0 \gamma^0 = \gamma^0 (\gamma^0)^2 \quad (56)$$

$$(\gamma^0)^2 = 1 \quad (57)$$

Here 1 is shorthand for the 4D identity matrix.

$$\bar{\gamma}^0 = \gamma^0 \quad (58)$$

Now the $\mu = i = 1, 2, 3$ case.

$$\gamma^i = \bar{\gamma}^i = \gamma^0 \gamma^{i\dagger} \gamma^0 \quad (59)$$

$$\gamma^{i\dagger} = -\gamma^i \quad (60)$$

$$\bar{\gamma}^i = -\gamma^0 \gamma^i \gamma^0 \quad (61)$$

$$\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu} \quad (62)$$

$$\{\gamma^i, \gamma^0\} = 2\eta^{i0} = 0 \implies \gamma^i \gamma^0 = -\gamma^0 \gamma^i \quad (63)$$

$$\bar{\gamma}^i = \gamma^i \gamma^0 \gamma^0 = \gamma^i (\gamma^0)^2 = \gamma^i \quad (64)$$

$$\bar{\gamma}^i = \gamma^i \quad (65)$$

Thus

$$\boxed{\bar{\gamma}^\mu = \gamma^\mu} \quad (66)$$

for $\mu = 0, 1, 2, 3$. QED.

2.2. $S^{\mu\nu}$

$$S^{\mu\nu} = \frac{i}{4} [\gamma^\mu, \gamma^\nu] \quad (67)$$

$$\bar{S}^{\mu\nu} = \gamma^0 (S^{\mu\nu})^\dagger \gamma^0 \quad (68)$$

$$S^{\mu\nu} = \frac{i}{4} (\gamma^\mu \gamma^\nu - \gamma^\nu \gamma^\mu) \quad (69)$$

$$\begin{aligned} (S^{\mu\nu})^\dagger &= -\frac{i}{4} (\gamma^\mu \gamma^\nu - \gamma^\nu \gamma^\mu)^\dagger = -\frac{i}{4} [(\gamma^\mu \gamma^\nu)^\dagger - (\gamma^\nu \gamma^\mu)^\dagger] \\ &= -\frac{i}{4} (\gamma^{\nu\dagger} \gamma^{\mu\dagger} - \gamma^{\mu\dagger} \gamma^{\nu\dagger}) \end{aligned} \quad (70)$$

$$(S^{\mu\nu})^\dagger = \frac{i}{4} (\gamma^{\mu\dagger} \gamma^{\nu\dagger} - \gamma^{\nu\dagger} \gamma^{\mu\dagger}) \quad (71)$$

$$\bar{S}^{\mu\nu} = \frac{i}{4} \gamma^0 (\gamma^{\mu\dagger} \gamma^{\nu\dagger} - \gamma^{\nu\dagger} \gamma^{\mu\dagger}) \gamma^0 \quad (72)$$

$$\bar{S}^{\mu\nu} = \frac{i}{4} (\gamma^0 \gamma^{\mu\dagger} \gamma^{\nu\dagger} \gamma^0 - \gamma^0 \gamma^{\nu\dagger} \gamma^{\mu\dagger} \gamma^0) \quad (73)$$

$$\bar{S}^{\mu\nu} = \frac{i}{4} (\gamma^0 \gamma^{\mu\dagger} \gamma^0 \gamma^0 \gamma^{\nu\dagger} \gamma^0 - \gamma^0 \gamma^{\nu\dagger} \gamma^0 \gamma^0 \gamma^{\mu\dagger} \gamma^0) \quad (74)$$

$$\bar{S}^{\mu\nu} = \frac{i}{4} (\bar{\gamma}^\mu \bar{\gamma}^\nu - \bar{\gamma}^\nu \bar{\gamma}^\mu) \quad (75)$$

$$\bar{S}^{\mu\nu} = \frac{i}{4} [\bar{\gamma}^\mu, \bar{\gamma}^\nu] \quad (76)$$

I use Eq. 66.

$$\bar{S}^{\mu\nu} = \frac{i}{4} [\gamma^\mu, \gamma^\nu] \quad (77)$$

$$\boxed{\bar{S}^{\mu\nu} = S^{\mu\nu}} \quad (78)$$

QED

2.3. $i\gamma_5$

$$\gamma_5 = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \quad (79)$$

Again every element is really a 2×2 matrix.

$$\overline{i\gamma_5} = \gamma^0 (i\gamma_5)^\dagger \gamma^0 \quad (80)$$

$$\overline{i\gamma_5} = -i\gamma^0 \gamma_5 \gamma^0 \quad (81)$$

I did the matrix multiplication in Mathematica.

$$\boxed{\overline{i\gamma_5} = i\gamma_5} \quad (82)$$

QED

2.4. $\gamma^\mu \gamma_5$

$$\overline{i\gamma_5 S^{\mu\nu}} = i\frac{i}{4}(\gamma^\mu \gamma^\nu \gamma_5 - \gamma^\nu \gamma^\mu \gamma_5) \quad (100)$$

$$\overline{\gamma^\mu \gamma_5} = \gamma^0 (\gamma^\mu \gamma_5)^\dagger \gamma^0 \quad (83)$$

$$\overline{i\gamma_5 S^{\mu\nu}} = i\frac{i}{4}(-\gamma^\mu \gamma_5 \gamma^\nu + \gamma^\nu \gamma_5 \gamma^\mu) \quad (101)$$

$$\overline{\gamma^\mu \gamma_5} = \gamma^0 \gamma_5 \gamma^{\mu\dagger} \gamma^0 \quad (84)$$

$$\overline{i\gamma_5 S^{\mu\nu}} = i\frac{i}{4}(\gamma_5 \gamma^\mu \gamma^\nu - \gamma_5 \gamma^\nu \gamma^\mu) \quad (102)$$

$$\overline{\gamma^\mu \gamma_5} = \gamma^0 \gamma_5 \gamma^0 \gamma^{\mu\dagger} \gamma^0 \quad (85)$$

$$\overline{i\gamma_5 S^{\mu\nu}} = i\gamma_5 \frac{i}{4}(\gamma^\mu \gamma^\nu - \gamma^\nu \gamma^\mu) \quad (103)$$

$$\overline{\gamma^\mu \gamma_5} = \gamma^0 \gamma_5 \gamma^0 \bar{\gamma}^\mu \quad (86)$$

$$\overline{\gamma^\mu \gamma_5} = \gamma^0 \gamma_5 \gamma^0 \gamma^\mu \quad (87)$$

$$\overline{i\gamma_5 S^{\mu\nu}} = i\gamma_5 \frac{i}{4}[\gamma^\mu, \gamma^\nu] \quad (104)$$

$$\{\gamma^5, \gamma^\mu\} = 0 \implies \gamma^5 \gamma^\mu = -\gamma^\mu \gamma^5 \quad (88)$$

$$\boxed{\overline{i\gamma_5 S^{\mu\nu}} = i\gamma_5 S^{\mu\nu}} \quad (105)$$

$$\overline{\gamma^\mu \gamma_5} = -\gamma_5 \gamma^0 \gamma^0 \gamma^\mu \quad (89)$$

QED

$$\overline{\gamma^\mu \gamma_5} = -\gamma_5 \gamma^\mu \quad (90)$$

$$\boxed{\overline{\gamma^\mu \gamma_5} = \gamma^\mu \gamma_5} \quad (91)$$

QED

2.5. $i\gamma_5 S^{\mu\nu}$

$$\overline{i\gamma_5 S^{\mu\nu}} = \gamma^0 (i\gamma_5 S^{\mu\nu})^\dagger \gamma^0 \quad (92)$$

$$\overline{i\gamma_5 S^{\mu\nu}} = \gamma^0 (S^{\mu\nu})^\dagger (i\gamma_5)^\dagger \gamma^0 \quad (93)$$

$$\overline{i\gamma_5 S^{\mu\nu}} = \gamma^0 (S^{\mu\nu})^\dagger \gamma^0 \gamma^0 (i\gamma_5)^\dagger \gamma^0 \quad (94)$$

$$\overline{i\gamma_5 S^{\mu\nu}} = \bar{S}^{\mu\nu} \overline{i\gamma_5} \quad (95)$$

I use Eqs. 78 and 82.

$$\overline{i\gamma_5 S^{\mu\nu}} = S^{\mu\nu} i\gamma_5 \quad (96)$$

$$\overline{i\gamma_5 S^{\mu\nu}} = iS^{\mu\nu} \gamma_5 \quad (97)$$

$$\overline{i\gamma_5 S^{\mu\nu}} = i\frac{i}{4}[\gamma^\mu, \gamma^\nu] \gamma_5 \quad (98)$$

$$\overline{i\gamma_5 S^{\mu\nu}} = i\frac{i}{4}(\gamma^\mu \gamma^\nu - \gamma^\nu \gamma^\mu) \gamma_5 \quad (99)$$